



Find the Loop Bug

Ontario Math C3.2 — read, find, and fix code

Name: _____

Date: _____

★ I can find the bug in a loop and fix it.

Coders fix bugs

A bug is a mistake in code. Coders read carefully to FIND the bug, then FIX it — like finding a wrong step in a set of directions.

A loop bug is usually one of two things: the wrong repeat number, or a wrong block inside. Help Bit find and fix them.

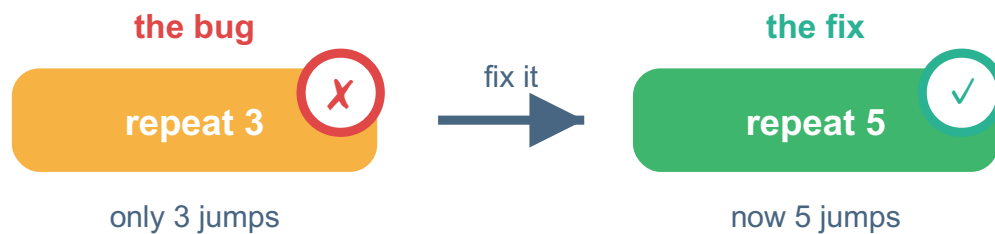
Bit should JUMP 5 times — but this loop is buggy

when green flag clicked

repeat 3

jump

Bit only jumps 3 times. The repeat number is wrong — it should be 5.



Find the wrong number, then fix it: repeat 3 → repeat 5.

1 Find the bug

Bit should jump 5 times, but the loop says repeat 3. What is wrong? What should the repeat number be? _____

2 Fix the inside

Bit should stamp red, blue. But the loop's inside is: stamp red, stamp red. Which block is wrong, and what should it be?

3 Fix the count

Bit should move 30 steps in total, 10 each time. The loop says repeat 2. What repeat number fixes it?

4 Challenge

This loop is repeat 4 with TWO "move" blocks inside. Bit moved 8 times but should move only 4. Is the bug the repeat number or the inside blocks? Fix it and explain.

Big idea

To debug a loop, first say what it **SHOULD** do, then read what it **DOES**. The bug is usually the repeat number or a block inside — change the one that is wrong.



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Quick facilitation notes & answer key

What this worksheet teaches

Students read a buggy loop, decide whether the bug is the repeat number or an inside block, and fix it so the code matches the goal. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). Compare the loop to the goal: is the repeat number wrong, or is an inside block wrong?
2. Do one together. Exercise 1 — Bit should jump 5 times but the loop says repeat 3; fix the repeat number.
3. Let them work. Students fix a wrong inside block (Ex 2), a wrong repeat (Ex 3), and decide which part is the bug (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — the repeat number is wrong; it should be repeat 5. Exercise 2 — the second block is wrong; change "stamp red" to "stamp blue".

Exercise 3 — repeat 3 ($3 \times 10 = 30$ steps). Exercise 4 (challenge) — the inside is the bug: there are two move blocks; remove one so repeat 4 gives 4 moves.

If students get stuck — and ways to adjust

Watch for: always blaming the repeat number. Sometimes an inside block is the bug.

Make it easier: ask first "is the count wrong, or is one of the actions wrong?"

Make it harder: ask them to name the bug type for each exercise.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child decides whether the repeat number or an inside block is wrong and fixes it to match the goal.